

Python Programming Lab: File Handling

Course: Python with Data Science

Topics: Text Processing, CSV Management, JSON Data, and Binary Operations

Part 1: Easy (Foundational Operations)

1. The Log Writer * Task: Create a program that writes your name and current course name to a file called `student_info.txt`.

- **Step 2:** Close the file, then reopen it in **append mode** ('a') to add a new line stating your favorite programming language.
- **Step 3:** Reopen the file in **read mode** ('r') and print the final contents to the console.

2. The Line Counter * Task: Manually create a text file named `notes.txt` with at least 5 lines of random text (e.g., a "To-Do" list).

- **Step 2:** Write a Python script that opens this file, iterates through it using a for loop, and prints the total number of lines found in the file.
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Part 2: Intermediate (Structured Data)

3. The Electronics Catalog (CSV) * Task: Create a list of lists named `product_catalog` containing exactly **10 rows** of data. Each row should include [Product Name, Category, Price].

- **Step 2:** Use the `csv` module and `writer.writerows()` to save this data into a file named `electronics.csv`.
- **Constraint:** Ensure you use the `newline=""` parameter in your `open()` function to prevent empty rows between data.

4. Phone Inventory (JSON) * Task: Define a list of dictionaries representing three different phones (e.g., `{"Brand": "Apple", "Model": "iPhone 15"}`).

- **Step 2:** Save this list into a file named `inventory.json` using `json.dump()`.
- **Step 3:** Write a script to load the data back and print only the **Model** names.

5. User Database (DictReader) * Task: Using a CSV file with headers Name and Age, write a script that uses csv.DictReader to process the file.

- **Output:** For every row, print: "User [Name] is [Age] years old."
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Part 3

6. The Image Cloner (Binary Mode) * Task: Write a script to duplicate an image file using binary modes.

- **Step 2:** Open an existing image (e.g., input.png) in **binary read mode** ('rb').
- **Step 3:** Open a new file named clone.png in **binary write mode** ('wb') and write the data from the first file into the second.

7. Dynamic CSV Header Mapper * Task: Create a dictionary representing a student:
student_data = {"ID": 101, "Name": "Sita", "Grade": "A"}.

- **Step 2:** Write a script that automatically extracts the **keys** from this dictionary to use as fieldnames for a csv.DictWriter.
- **Step 3:** Save the data into students.csv.